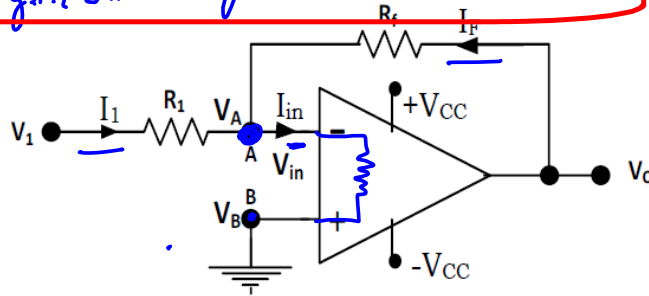


- * Opamp'in tencine alim almaz:
- * Opamp'in girislerindeki gerilimler birbirine esittir.

$$I_{in} = 0$$



$$V_B = 0V \Rightarrow V_A = 0V$$

KAY'a göre bir noktaya gelen dumanların toplamı, giden dumanların toplamına eşittir.

$$I_1 + I_f = I_{in}$$

$$I_1 = -I_f$$

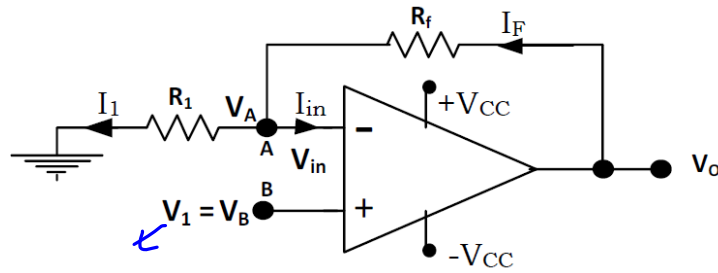
$$\frac{V_1 - V_A}{R_1} + \frac{V_0 - V_A}{R_f} = 0$$

$$\frac{V_1}{R_1} = -\frac{V_0}{R_f}$$

Giris gerilimi $\leftarrow \frac{V_1}{R_1}$ $\rightarrow$ Çikis gerilimi $\frac{V_0}{R_f}$

$$Kazan = \frac{\text{Çikis}}{\text{Giris}}$$

$$A_v = \frac{V_0}{V_1} = -\frac{R_f}{R_1}$$



$$I_1 + I_{in} = I_f$$

$$\frac{V_A - 0V}{R_1} = \frac{V_O - V_A}{R_f}$$

$$V_A = V_B = V_1$$

$$\frac{V_1}{R_1} = \frac{V_O - V_1}{R_f} \quad \left(\frac{V_O}{V_1} \right)$$

$$\frac{V_1}{R_1} = \frac{V_O}{R_f} - \frac{V_1}{R_f}$$

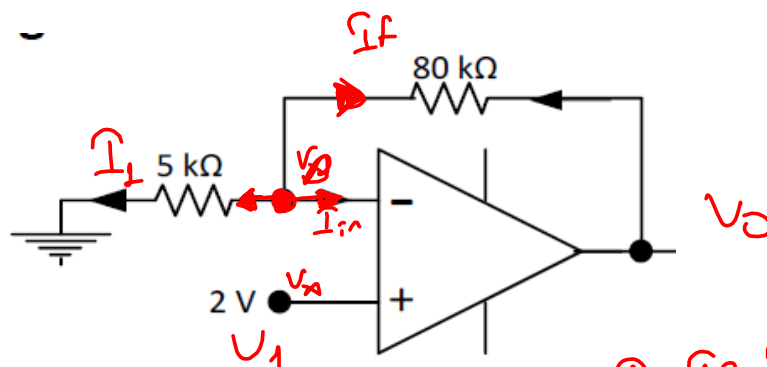
$$\frac{V_1}{R_1} + \frac{V_1}{R_f} = \frac{V_O}{R_f}$$

$$V_1 \left(\frac{1}{R_1} + \frac{1}{R_f} \right) = \frac{V_O}{R_f}$$

$$\frac{V_O}{V_1} = \frac{R_f}{R_1} + 1$$

$$A_v = \frac{V_O}{V_1} = \left(1 + \frac{R_f}{R_1} \right)$$

$$V_O = V_1 \cdot \left(1 + \frac{R_f}{R_1} \right)$$



$$V_B = V_A = V_1 = \underline{2V}$$

$$I_1 + I_f + I_{in} = 0$$

$$\downarrow$$

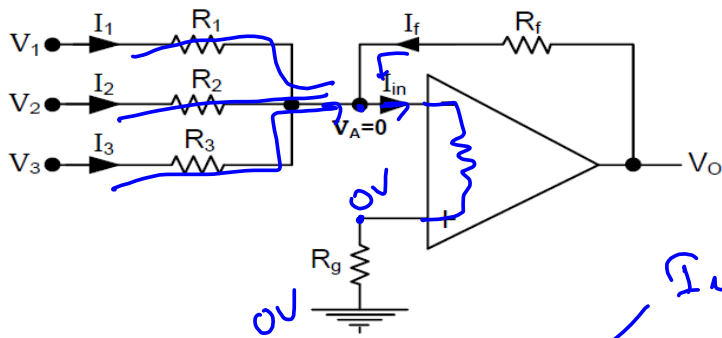
$$\frac{V_B - 0V}{5k} + \frac{V_B - V_o}{80k} + 0 = 0$$

$$\frac{2V}{5k} + \frac{2V - V_o}{80k}$$

$$\frac{2V}{5k} + \frac{2V}{80k} = \frac{V_o}{80k}$$

$$V_o = 80k \cdot \left(\frac{2V}{5k} + \frac{2V}{80k} \right)$$

$$\begin{aligned} & \underline{16.2V + 2V} \\ & \underline{2V(18+1)} \\ & \underline{V_1 \quad \text{loopback}} \\ & \underline{32V} \end{aligned}$$

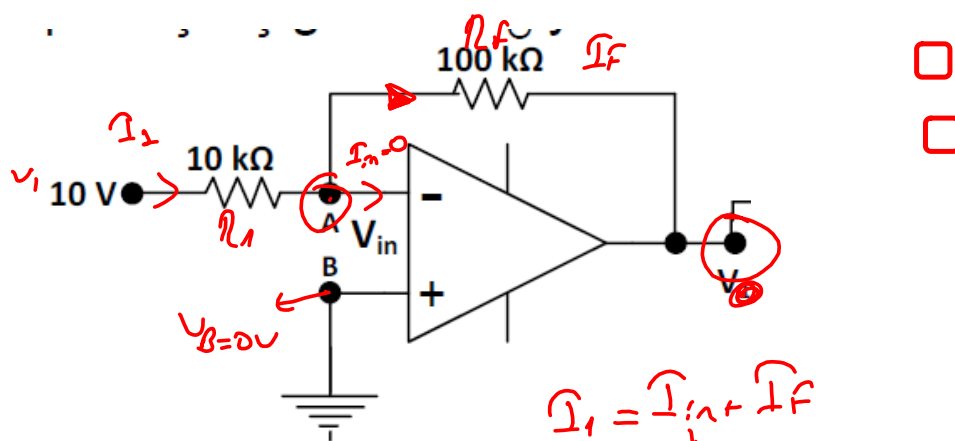


$$I_1 + I_2 + I_3 + I_f = I_{in} \downarrow \uparrow 0$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_o}{R_f}$$

Eğer $R_1 = R_2 = R_3 = R_f$ ise

$$V_o = -(V_1 + V_2 + V_3) \text{ olur.}$$



$$I_1 = I_{in} + I_F$$

$$I_1 = I_F$$

$$\frac{V_1 - V_A}{R_1} = \frac{V_A - V_o}{R_F}$$

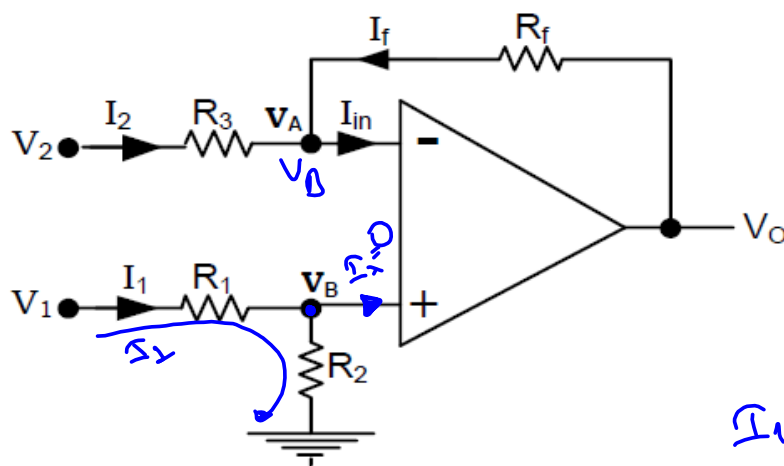
$$\frac{10\text{ V} - 0\text{ V}}{10\text{ k}} = \frac{0 - V_o}{100\text{ k}}$$

$$\downarrow$$

$$\frac{1\text{ mA}}{100\text{ k}}$$

$$-V_o = 100\text{ V}$$

$$V_o = -100\text{ V}$$



$$I_{in} = 0$$

$$V_A = V_B$$

$$V_B = \frac{V_1}{R_1 + R_2} \cdot R_2$$

$$I_1 = \frac{V_1}{R_1 + R_2} \quad V_0 = I_1 \cdot R_2$$

$$I_2 + I_f = I_{in}$$

$$\frac{V_2 - V_B}{R_3} + \frac{V_0 - V_B}{R_f} = 0$$

$$R_1 = R_2 \quad \text{ve} \quad R_3 = R_f \quad \text{olursa}$$

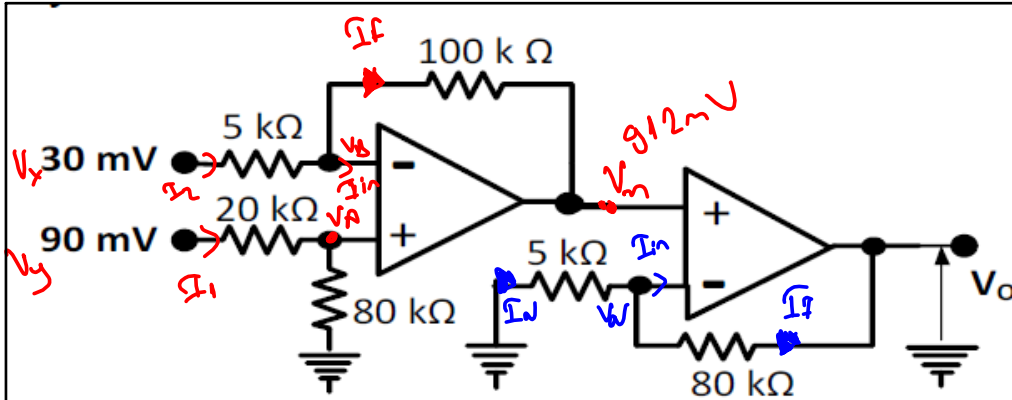
$$R_3 = R_f \quad \text{ise}$$

$$V_2 - 2V_B = -V_0 \quad R_1 = R_2 \Rightarrow V_B = \frac{V_1}{2}$$

$$V_0 = -\left(V_2 - 2 \cdot \frac{V_1}{2}\right)$$

$$V_0 = -(V_2 - V_1)$$

$$V_0 = V_1 - V_2$$



$$V_A = \frac{V_y}{20k + 80k} \cdot 80k = \frac{90 \cdot 10^{-3} \cdot 80k}{100k} = 72mV \Rightarrow V_B = 72mV$$

$$V_N = V_m = 912mV$$

$$I_2 = I_{in} + I_f$$

$$I_N = I_{in} + I_f$$

$$\frac{V_x - V_B}{5k} = 0 + \frac{V_B - V_m}{100k}$$

$$\frac{0 - 912mV}{5k} = \frac{912mV - V_o}{80k}$$

$$\frac{30mV - 72mV}{5k} = \frac{72mV - V_m}{\frac{100k}{20}}$$

$$V_o = 17 \cdot (912mV) + 912mV$$

$$V_o = 17 \cdot 912mV = 15,504mV$$

$$(-42mV) \cdot 20 = 72mV - V_m$$

$$V_m = 840mV + 72mV$$

$$V_m = 912mV$$